

AI UNLEASHES CHAOS

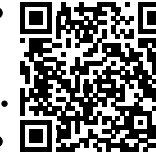
Lorenz attractor -- circuit by Paul Horowitz
PCB by Jason Gallicchio and Claude Opus 5 Max | 2026-09-15 rev A



$$\begin{aligned}dx/dt &= s (y - x) \\dy/dt &= r x - y - x z \\dz/dt &= x y - b z\end{aligned}$$

$s = 10 \quad r = 21 \text{ to } 37 \quad b = 8/3$

put x and z on X-Y to see this



this project, and how
it was made
[github.com/gallicchio
/aLunleashes_chaos](https://github.com/gallicchio/aLunleashes_chaos)

Every term weighs 1 MEG / R.
s is R1 = R2 = 100k, and b is R7 = 374k.
r is R3 = 27k plus the knob RV1, which sweeps r from 21 to 37.
The MPY634s form A*B/10, and both minus signs come from swapped inputs.
• Signals are 0.1 V per unit, and the time scale is tau = 1 MEG x C.



Paul's original circuit
[seti.harvard.edu/unusual_stuff
/misc/lorenz.htm](https://seti.harvard.edu/unusual_stuff/misc/lorenz.htm)

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